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ADVERSARIALLY ROBUST PHISHING URL DETECTION USING FINE-TUNED DISTILBERT AND RETRIEVAL-AUGMENTED EXPLANATIONS

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INTRODUCTION & PROBLEM STATEMENT

Introduction

- Phishing is a cyberattack in which attackers use deceptive URLs to trick users into visiting fraudulent websites.
- These URLs often imitate trusted services such as banks, email providers, and popular websites.
- Attackers modify URLs using character substitutions, misleading subdomains, and other techniques to avoid detection.
- Machine learning can identify suspicious URL patterns before a user opens a website.

Problem Statement



Traditional phishing detection models may perform well on normal URLs but can struggle when URLs are deliberately modified to bypass detection.

Our goal:



Develop a phishing URL detection framework that combines fine-tuned DistilBERT, adversarial URL testing, retrieval-based score adjustment, and AI-generated explanations.

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OBJECTIVES & EXISTING RESEARCH

Project Objectives

1.Develop a machine learning-based phishing URL detection system → 2.Fine-tune DistilBERT to classify URLs as phishing or legitimate. → 3.Evaluate model performance against five types of URL mutations. → 4.Integrate FAISS-based retrieval for additional threat-pattern information. → 5.Generate understandable explanations using Flan-T5.

Existing Research & Research Gap →

Research	Main approach
Sahingoz et al. (2019)	URL-based machine learning
Li et al. (2025)	URLBERT and adversarial pre-training
Said et al. (2024)	CNN with self-attention
Rashid et al. (2024)	Cross-dataset generalisation
Kulkarni et al. (2025)	Adversarial phishing webpage detection

Identified Research Gap

Our work investigates the combination of URL mutation-based evaluation, transformer classification, retrieval-based score adjustment, and separate explanation generation within one framework.

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Proposed Methodology

The proposed system follows a multi-stage approach:

Stage 1: Dataset Collection
- Collect phishing and legitimate URLs from two public datasets.
- Merge and clean the collected data.

Stage 2: Data Preprocessing
- Remove duplicates and conflicting entries.
- Filter invalid URLs.
- Balance the dataset.
- Split data into training, validation, and testing sets.

Stage 3: Adversarial Augmentation
- Generate modified URLs using five mutation techniques.
- Add mutated samples to the training data.

Stage 4: Model Training
- Train a TF-IDF and Logistic Regression baseline.
- Fine-tune DistilBERT on clean and augmented datasets.

Stage 5: Retrieval & Explanation
- Retrieve relevant threat patterns using Sentence-BERT and FAISS.
- Apply additional rules for score adjustment.
- Generate explanations using Flan-T5.

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Dataset & Adversarial URL Mutations

Dataset Information

Two public datasets were used:

- kmack/Phishing_urls
https://huggingface.co/datasets/kmack/Phishing_urls

- ealvaradob/phishing-dataset
https://huggingface.co/datasets/ealvaradob/phishing-dataset

Five Adversarial Mutation Techniques

1)Homoglyph
Replacing characters with visually similar characters

2)Subdomain Spoofing
Adding a trusted brand name to a misleading subdomain

3)Typosquatting
Changing or swapping characters

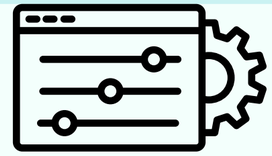
4)URL Padding
Adding suspicious-looking query parameters

5)Percent Encoding
Encoding characters within the URL

Description	Number of URLs
Candidate pool	4000
Phishing URLs	1000
Legitimate URLs	1000
Training set	1400
Validation set	300
Testing set	300
Augmented training set	2310
Adversarial test set	300

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System Architecture



Input: Phishing & Legitimate URLs

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Data Preprocessing & Cleaning

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Adversarial URL Augmentation
Five mutation techniques:

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classification Models

M1
TF-IDF + Logistic Regression

M2
Clean-trained DistilBERT

M3
Augmentation-trained DistilBERT

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Hybrid Retrieval & Score Adjustment (M4/M5)
Sentence-BERT + FAISS + Rules

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Flan-T5 Explanation Generation
Separate explanation path; does not change the verdict

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Final Output
Phishing / Legitimate + Confidence + Explanation

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Models & Technologies Used

Model	Configuration	Purpose
M1	TF-IDF + Logistic Regression	Baseline model
M2	Clean-trained DistilBERT	Transformer baseline
M3	Augmentation-trained DistilBERT	Effect of adversarial augmentation
M4	M3 + FAISS retrieval	Retrieval-based score adjustment
M5	M3 + Retrieval + Rules	Full hybrid framework

Additional Technologies

- **DistilBERT: Transformer-based URL classification.**
- **Sentence-BERT: Converts threat information into vector embeddings.**
- **FAISS: Performs similarity-based retrieval.**
- **Flan-T5: Generates understandable explanations.**
- **Python & PyTorch: Model implementation and training.**
- **Scikit-learn: Baseline model and evaluation.**
- **Gradio: User interface for URL scanning and testing.**

Important Design Decision

The classification model makes the prediction. Retrieval and rules adjust the score in M4/M5, while Flan-T5 generates a separate explanation. The explanation model does not determine the final classification.

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Experimental Setup & Evaluation Metrics

Experimental Environment

- Programming Language: Python
- Platform: Google Colab
- Hardware: NVIDIA T4 GPU
- Dataset split: 70:15:15
- Random seed: 42
- DistilBERT maximum sequence length: 128 tokens
- Learning rate: 3×10^{-5}
- Batch size: 8
- Training epochs: 2

Clean vs Adversarial Testing

The models were evaluated on:

. 300 clean test URLs.

300 adversarial test URLs, consisting of 150 mutated phishing URLs and 150 unchanged legitimate URLs.

Metric	Purpose
Accuracy	Overall correct predictions
Precision	Correct phishing predictions among predicted phishing URLs
Recall	Phishing URLs correctly identified
F1-score	Balance between precision and recall
ROC-AUC	Ability to distinguish between classes
FPR	Percentage of legitimate URLs incorrectly flagged

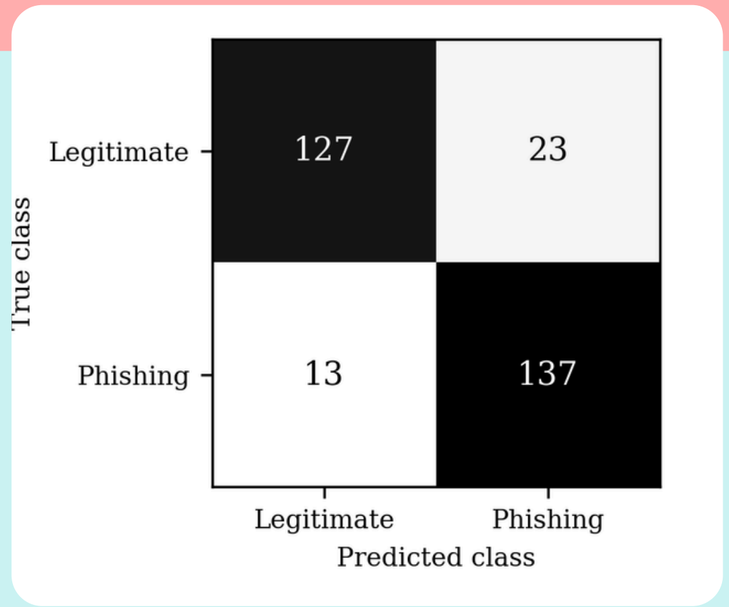
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Results & Performance Analysis

Model	Clean Accuracy	Clean F1	Adversarial Accuracy	Adversarial F1
M1	85.00%	0.8454	86.33%	0.8610
M2	89.33%	0.8919	92.67%	0.9281
M3	87.00%	0.8696	88.00%	0.8808
M4	87.00%	0.8696	88.67%	0.8882
M5	85.67%	0.8581	88.00%	0.8839

Key Observations

- M2 achieved the highest F1-score on both clean and adversarial test sets.
- Adversarial augmentation did not outperform the clean-trained DistilBERT model.
- M4 improved adversarial F1 over M3 from 0.8808 to 0.8882.
- M5 achieved a mutated-test recall of 91.33%.
- The additional rules in M5 increased false positives compared with M3.
- M4 confusion matrix;



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Conclusion, Limitations & Future Work

Conclusion

- Developed a layered phishing URL detection prototype using DistilBERT and retrieval-based techniques.
- Evaluated five model configurations under clean and synthetic adversarial conditions.
- M2, the clean-trained DistilBERT model, achieved the best overall F1-score.
- Retrieval-based adjustment provided a small improvement over M3 on the adversarial test set.
- Flan-T5 provided separate natural-language explanations for model predictions.

Limitations

- Only 2,000 balanced URLs were used in the final dataset.
- Evaluation was performed using a single seeded split.
- The same mutation families were used for augmentation and testing.
- The threat knowledge base contains only four entries.
- Explanation quality and inference latency were not formally measured.

Future Work

- Test on larger and more diverse datasets.
- Evaluate against unseen and combined URL mutations.
- Perform repeated experiments using multiple random seeds.
- Expand the threat knowledge base.
- Evaluate explanation quality through human reviewers.
- Measure inference time and deployment performance.



THANK YOU